

Giant Pressure-Induced Spectral Shift in Cyan-Emitting Eu^{2+} -Activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ Microspheres for Ultrasensitive Visual Manometry

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To address the unsatisfactory pressure sensitivity of luminescent manometers, Eu^{2+} -activated supersensitive microspheres operating in the visible range are developed. A series of Eu^{2+} -doped $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ materials are synthesized as microspheres, and their structural and spectroscopic properties are studied theoretically and experimentally. Excited at 350 nm, the samples emit a bright cyan luminescence at ambient conditions that, upon pressure, changes to green emission and finally to yellow light above 7 GPa. Most importantly, a huge red-shift of the emission band from 497.3 to 568.8 nm is observed as the pressure increases, leading to an ultrahigh-pressure sensitivity of 9.69 nm/GPa, which is the highest sensitivity ever reported. The designed microspheres with polychromatic emissions and high-pressure sensitivity are suitable for visual optical pressure sensing, and the applied strategy provides some important guidelines for the development of new optical manometers, allowing pressure monitoring with unprecedented accuracy.

diamond anvil cell (DAC) system is widely used, in which optical pressure monitoring is available due to the high transparency of diamonds in a broad spectral range, i.e., from ultraviolet (UV), via visible to near-infrared (NIR) light.^[7,8] In order to monitor the static pressure in a DAC chamber, luminescent materials, in which lanthanide or transition metal ions act as activators, are often used by analyzing their pressure-dependent spectral characteristics, such as band position, full width at half maximum (FWHM), emission decay time and band ratio.^[8–10] So far, the analysis of the pressure-induced band shift of Cr^{3+} in $\text{Al}_2\text{O}_3:\text{Cr}^{3+}$ ruby, first reported by Forman et al., is the most commonly used pressure gauge, with a sensitivity of $d\lambda/dP = 0.365 \text{ nm/GPa}$.^[11,12]

Evidently, the pressure sensitivity of $\text{Al}_2\text{O}_3:\text{Cr}^{3+}$ ruby is not high enough to catch up with the requirement of technological revolution, i.e., higher sensitivity and better accuracy. Thus, researchers had developed other luminescent materials for pressure sensing, such as $\text{BaLi}_2\text{Al}_2\text{Si}_2\text{N}_6:\text{Eu}^{2+}$ ($d\lambda/dP = 1.58 \text{ nm/GPa}$),^[13] $\text{YVO}_4:\text{Yb}^{3+}/\text{Er}^{3+}$ ($d\lambda/dP = 1.77 \text{ nm/GPa}$),^[14] $\text{Ca}_2\text{Gd}_8\text{Si}_6\text{O}_{26}:\text{Ce}^{3+}$ ($d\lambda/dP = 3.00 \text{ nm/GPa}$),^[15] $\text{Ca}_9\text{NaZn}(\text{PO}_4)_7:\text{Eu}^{2+}$ ($d\lambda/dP = 5.21 \text{ nm/GPa}$),^[16] $\text{Mg}_2\text{Gd}_8(\text{SiO}_4)_6\text{O}_2:\text{Ce}^{3+}/\text{Mn}^{2+}$ ($d\lambda/dP = 1.85 \text{ nm/GPa}$),^[17] etc. Although the pressure sensitivity has been enhanced, some of

1. Introduction

Pressure, a fundamental thermodynamic parameter, exhibits a significant impact on various fields, such as scientific research, industrial manufacturing, construction facilities, etc.^[1–3] When materials undergo a compression process triggered by high-pressure conditions, some unique phenomena may occur, such as phase transitions, formation of new compounds, defects generation, local symmetry modification, and so forth.^[4–6] Nowadays, to apply static high-pressure in the laboratory, the

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DOI: 10.1002/adfm.202214663

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them are only suitable for measuring very high-pressures, i.e., they do not exhibit linear/monotonous spectral shift within the relatively low-pressure regime. That is why, in the “low-pressure range” (below ≈ 10 GPa), where organic compounds and various soft materials can undergo several phase transitions, there is an “empty” pressure monitoring window that requires highly sensitivity optical manometers. Hence, development of modern high-sensitivity luminescent manometers that can operate in the mentioned low-pressure range is very important.

Luminescent materials doped with rare-earth ions, which can emit many emission lines ranging from UV to NIR light, have attracted considerable attention, as they can be employed in various fields such as contactless sensing, solid-state lighting, anti-counterfeiting, etc.^[18–21] Compared with other rare-earth ions, Eu^{2+} is considered as an excellent activator owing to its intense luminescence arising from the parity-allowed $5d \rightarrow 4f$ transition, in which $5d$ level is bare and its energy is sensitive to the chemical character of the local environment in the host lattices.^[22] In general, the energies of the emission bands of Eu^{2+} -activated compounds are strongly dependent on the covalency and crystal-field strength.^[23] Notably, high-pressure can shorten the interatomic distances, which favors the increase of the overlap between neighboring orbitals, leading to an enhancement of the crystal-field strength and covalency.^[24] Thereby, the luminescence properties of Eu^{2+} -activated compounds can be manipulated by applying external pressure, making them potential candidates for optical manometry. The luminescent materials developed, in addition to being sensitive to high-pressure, must also have good optical performance, i.e., stable and intense luminescence, as well as emission and excitation within the spectral ranges of the commonly used detectors and excitation light sources. Since the chemical environment of host lattices influences the luminescence behavior of Eu^{2+} ions, the selection of a suitable host is the most convenient way to achieve this goal. Recently, enormous interest in chlorosilicates has been aroused due to their amazing features, such as low preparation temperature, excellent chemical and physical properties etc., and many phosphors, such as $\text{Ca}_2\text{SiO}_3\text{Cl}_2:\text{Eu}^{2+}$, $\text{Ca}_9\text{Zn}(\text{SiO}_4)_4\text{Cl}_2:\text{Ce}^{3+}/\text{Mn}^{2+}$, and $\text{Ca}_3\text{SiO}_4\text{Cl}_2:\text{Eu}^{2+}$,^[25–27] have been developed for lighting applications. As a member of chlorosilicates, $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$, first reported by Jing et al. in 2001,^[28] with tetragonal phase, was found to be a good luminescent host for rare-earth ions. Moreover, it was found that the $\text{Ce}^{3+}/\text{Tb}^{3+}$ -codoped $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ phosphors can emit highly-efficient controllable color emissions, indicating that the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ is a good luminescent host.^[29] However, to our knowledge, the investigation on the luminescence properties of Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres under high-pressure is still lacking.

Accordingly, the series of Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres with intense cyan emissions were prepared by high-temperature solid-state reaction method. The crystal structure, surface performance, and luminescence characteristics of the resultant microspheres have been analyzed in detail. By analyzing the pressure-induced red-shift of emission band, the studied samples possess splendid pressure sensing capacity, with a sensitivity reaching 9.69 nm/GPa. Moreover, with increasing pressure, their emission colors change significantly at extreme pressure values. The stability of the phase structure under high-pressure was studied by density functional theory

(DFT) calculations under periodic boundary conditions, correlating the computational results to vibrational spectra and XRD data at varying pressure. These achievements imply that the highly efficient and splendidly sensitive Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres are promising candidates for modern optical manometry.

2. Results and Discussion

2.1. Properties at Ambient Conditions

The unit cell parameter of the pure $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ crystal structure at ambient conditions are: $a = b = 11.201$ Å; $c = 9.584$ Å; $V = 1202.39$ Å³. To explore the influence of the Eu^{2+} doping on the phase structure of the final products, the X-ray diffraction (XRD) profiles of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$ ($0.005 \leq x \leq 0.12$) microspheres were examined (see Figure 1a). Evidently, the diffraction profiles are hardly affected by Eu^{2+} content and agree well with the standard reference pattern of the pure $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ (ICSD#1 535 962), implying that the resultant microspheres have a single tetragonal phase. Due to the similarity of ionic radii of Eu^{2+} and Sr^{2+} ($\text{Sr}^{2+} = 1.26$ Å, $\text{Eu}^{2+} = 1.25$ Å), the Eu^{2+} ion readily substitutes for Sr^{2+} one in the crystalline structure of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$. In order to deepen the crystal structure of the resulting samples, Rietveld structure refinements of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$ microspheres with the dopant contents of 4 and 12 mol% were carried out, as shown in Figure 1b,c, respectively. Clearly, the calculated diffraction profiles correspond well with the recorded XRD patterns, further indicating that the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres are successfully synthesized. Note that with increasing Eu^{2+} content, a slight decrease in the lattice parameters is observed (see Table S1, Supporting Information), which is due to the substitution of larger Sr^{2+} ions by smaller Eu^{2+} ones. Figure 1d shows the structural 3D representation of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ unit cell, depicting its tetragonal crystal structure with $I4/m$ (87) space group. As presented, four $[\text{SiO}_4]$ tetrahedrons share corners to form a four-member ring with a single Sr site, where the Sr^{2+} cation is coordinated by four oxygen and four chloride anions.^[29] Thus, when Eu^{2+} ion enters into the Sr^{2+} site in the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ host lattices, only one emitting centre can be generated, which will be discussed in the following section. To reveal the surface characteristics of the developed microspheres, the representative Fourier transform infrared (FT-IR) spectrum of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres were recorded (see Figure 1e). The characteristic absorption bands of the $[\text{SiO}_4]$ unit are located at ≈ 1007 , 912, 633, and 488 cm^{-1} , and correspond to antisymmetric stretching (ν_3), symmetric stretching (ν_1), symmetric bending (ν_2), and antisymmetric bending (ν_4) vibrations, respectively.

In order to reveal the influence of Eu^{2+} content on the morphology behavior of synthesized samples, the scanning electron microscope (SEM) images of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$ compounds were measured and the corresponding results are displayed in Figure 2a–d. As shown, the synthesized samples consist of micron-sized spherical particles. In particular, when $x = 0.005$, 0.04, 0.08, and 0.02, the average particle sizes of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$ compounds are 5.71, 5.79, 5.77, and

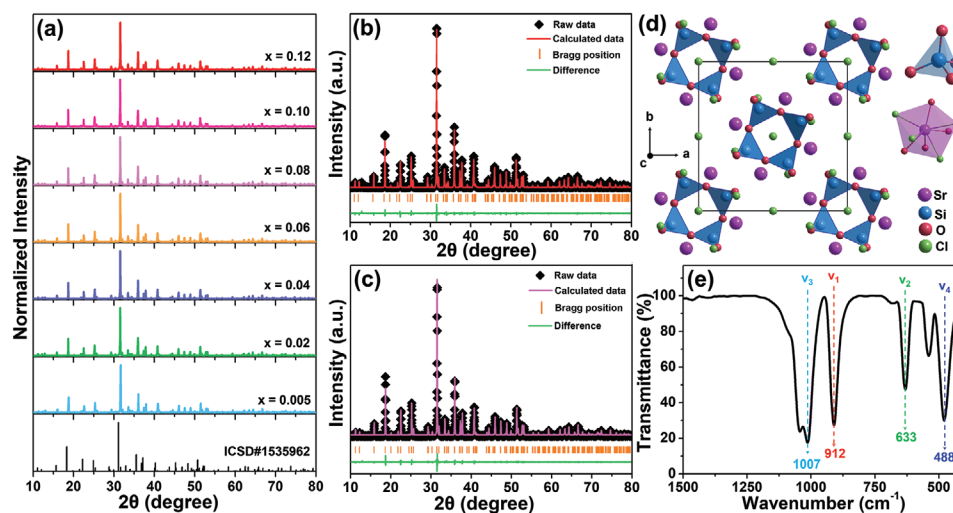


Figure 1. a) XRD patterns of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:xEu^{2+}$ ($0.005 \leq x \leq 0.12$) microspheres recorded at ambient conditions, using $\text{Cu K}\alpha 1$ radiation ($\lambda = 1.5406 \text{ \AA}$). Rietveld refinements of the b) $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32Eu^{2+}$ and c) $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.96Eu^{2+}$ microspheres. d) 3D representation of the crystal structure of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ unit cell. e) FT-IR spectrum of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32Eu^{2+}$ microspheres.

5.76 μm , respectively, as shown in Figure S1 (Supporting Information). Thus, it is clear that the morphology of the resultant microspheres is hardly affected by the Eu^{2+} content. Furthermore, the elemental compositions of the studied samples were verified by energy-dispersive X-ray spectroscopy (EDS) analysis

(see Figure S2, Supporting Information), where they are uniformly distributed throughout the microspheres, as displayed in Figure 2e–i.

To evaluate the luminescence properties of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:xEu^{2+}$ microspheres, their excitation and

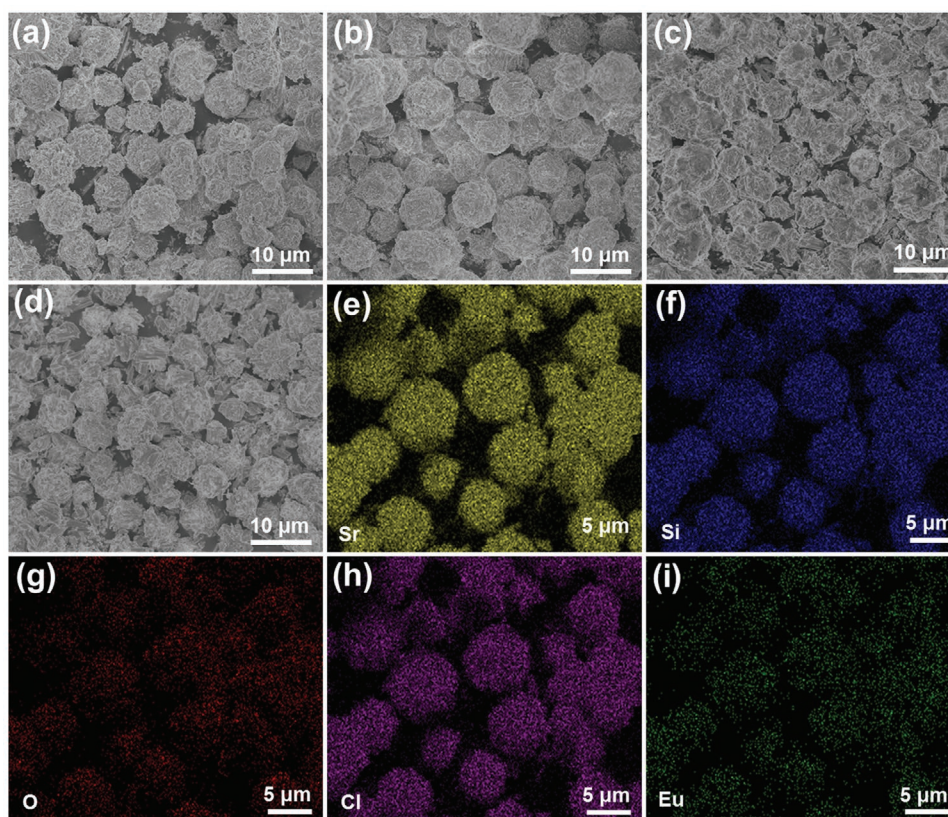


Figure 2. SEM graphs of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:xEu^{2+}$ microspheres with the doping contents of a) $x = 0.005$, b) $x = 0.04$, c) $x = 0.08$, d) $x = 0.12$. e–i) Elemental mapping of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32Eu^{2+}$ microspheres.

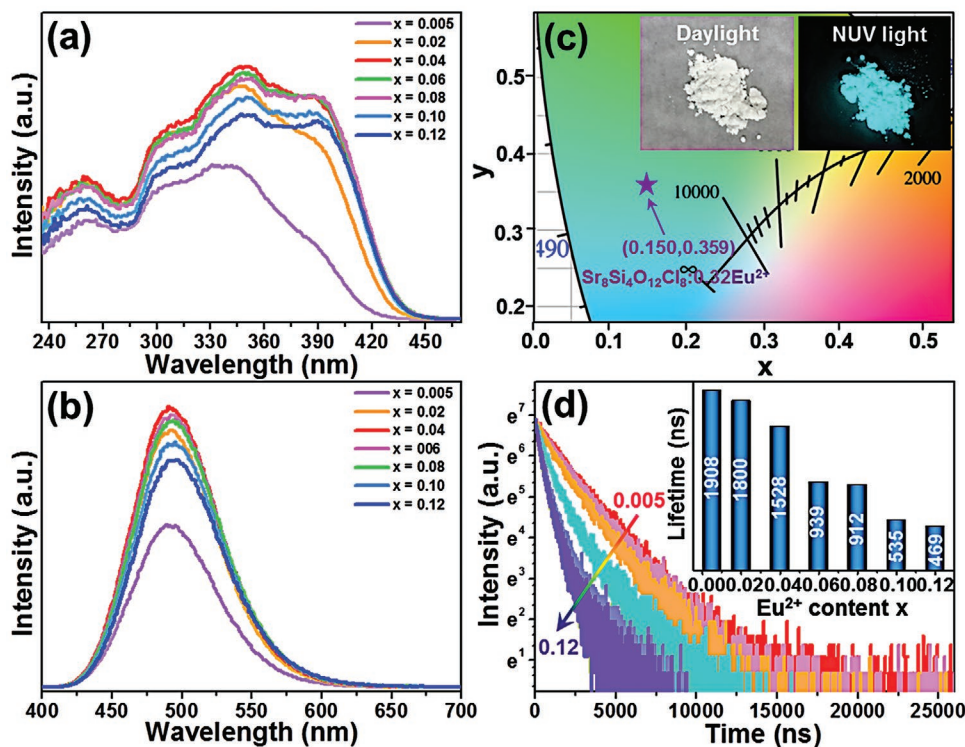


Figure 3. a) Excitation and b) emission spectra of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:\text{x}\text{Eu}^{2+}$ ($0.005 \leq x \leq 0.12$) microspheres. c) CIE diagram of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres and the luminescence photos taken under daylight and NUV light. d) Decay curves of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:\text{x}\text{Eu}^{2+}$ microspheres. Inset of (d) shows the determined luminescence decay times of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:\text{x}\text{Eu}^{2+}$ microspheres as a function of the Eu^{2+} content.

emission spectra were measured, as shown in Figure 3a, respectively. Monitored at 492 nm, the excitation spectra (see Figure 3a) consist of broad bands ranging from 236 to 450 nm, originating from the $4f \rightarrow 5d$ transition of Eu^{2+} ions, with the strongest intensity at 350 nm. The observed bands are split due to the crystal-field interaction between the lanthanide ion and its ligands. After irradiation at 350 nm, all the synthesized microspheres exhibit an intense broad emission band centered at ≈ 492 nm, which is assigned to the $5d \rightarrow 4f$ transition of Eu^{2+} ion (see Figure 3b).^[13,30] The emission band is symmetric, verifying the existence of a single emitting center for the Eu^{2+} ions in the host lattice. Moreover, the emission intensity is highly dependent on the Eu^{2+} content, and the optimal doping concentration, with the highest emission intensity, is found at $x = 0.04$. When the Eu^{2+} content is >4 mol%, the concentration quenching effect takes place, due to the nonradiative cross-relaxation energy transfer processes between Eu^{2+} ions, which results in the decrease of emission intensity. The detailed discussion of the concentration quenching mechanisms can be found in the Supporting Information data, where the plot of $\log(I/x)$ versus $\log(x)$ is shown in Figure S3 (Supporting Information). Figure 3c illustrates the color coordinates of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres determined from the corresponding emission spectrum. As shown, the prepared powder is white (in daylight) and can emit bright cyan emission with the color coordinates of (0.150, 0.359) under near-UV (NUV) irradiation, which highlights that the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$

microspheres can be used in lighting market as cyan-emitting components.

In order to understand the excited state dynamics of Eu^{2+} in $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ host lattice, the luminescence decay curves of the resulting microspheres were measured (Figure 3d), at $\lambda_{\text{ex}} = 372$ nm and $\lambda_{\text{em}} = 492$ nm. All the recorded profiles were fit to the following single exponential function:

$$I = I_0 + A \exp(-t/\tau) \quad (1)$$

where I_0 and I are the luminescence intensities at $t = 0$ and t , respectively; A is an amplitude and τ is the decay time. The single exponential behavior of the recorded decays confirms the presence of a single emitting site for Eu^{2+} ions in the studied materials. The lifetime of Eu^{2+} in $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ host lattices was determined to be 1908, 1800, 1528, 939, 912, 535, and 469 ns, respectively, when $x = 0.005, 0.02, 0.04, 0.06, 0.08, 0.10$, and 0.12 , as shown in the inset of Figure 3d. The average distance between neighboring Eu^{2+} ions decreases as the doping content increases, which favors the non-radiative energy transfer processes between Eu^{2+} ions and shortens the lifetime.^[31] In addition, the decreased decay time also verifies the existence of concentration quenching in the studied samples,^[15] which is in agreement with the results deduced from the emission spectra. In terms of practical applications, luminescent materials also need to possess high luminescence efficiency. In this case, the overall quantum yield, measured in a spherical integrator, for the optimal sample ($x = 0.04$), is $80 \pm 1\%$.

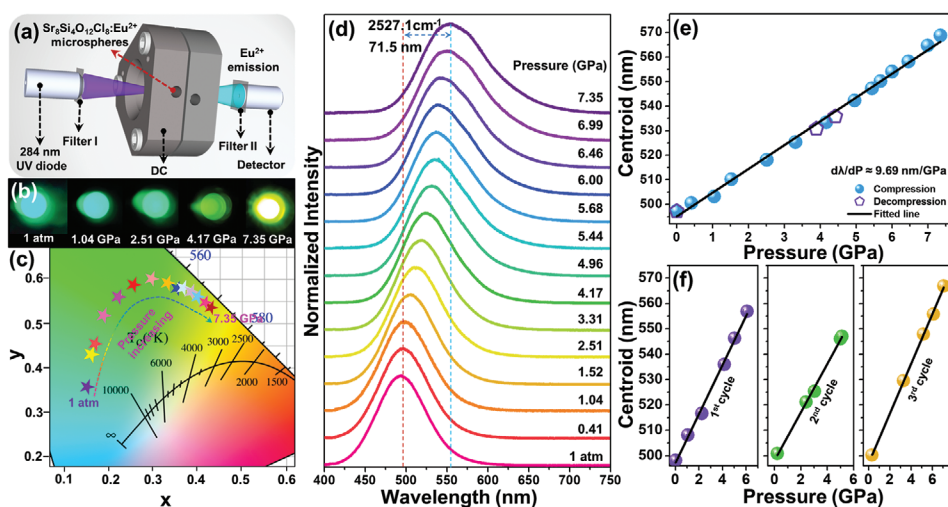


Figure 4. a) Experimental setup scheme for monitoring the pressure-dependent luminescence properties of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres. b) Digital images of the sample photoluminescence in a DAC, taken at various pressure values under UV light excitation (280 nm). c) CIE chromaticity diagram for the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres as a function of pressure. d) Normalized emission spectra of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres as a function of pressure; $\lambda_{\text{exc}} = 280 \text{ nm}$. e) Determined emission band centroid of Eu^{2+} as a function of pressure (0.0001–7.35 GPa). f) Cycling experiments, showing Eu^{2+} emission band centroid as a function of pressure, within three individual runs.

2.2. High-Pressure Sensing Performance

The featured emissions of Eu^{2+} originate from the transition of $5d \rightarrow 4f$ transitions, in which the excited $5d$ state is very sensitive to the surrounding chemical environment (crystal-field and nephelauxetic effects), leading to tunable spectroscopic properties by manipulating internal (e.g., structure or dopant ions) or external conditions, such as pressure and temperature.^[23] To analyze the influence of pressure on the luminescence properties of the synthesized materials, as well as explore their feasibility in optical manometry, the pressure-dependent emission spectra of the selected $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres were measured in the range of 0.0001 GPa (1 atm) to 7.35 GPa. **Figure 4a** shows the high-pressure measurement setup, in which a focused UV diode (280 nm) was used as the excitation source. Optical images taken at various pressure conditions under UV irradiation are presented in **Figure 4b**, where significant color-tuning luminescence, i.e., from cyan to green, and finally to yellow, is realized by adjusting the pressure values. **Figure 4c** shows the determined CIE coordinates for the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres at different pressures, which are estimated from the emission spectra under compression of the material (see **Figure 4d**). As pressure increases from ambient conditions up to 7.35 GPa, the CIE coordinates are shifted from (0.150, 0.402) to (0.430, 0.536), as shown in Table S2, Supporting Information, which agrees well with the observed visual changes, i.e., digital images shown in **Figure 4b**. Such a significant luminescence color change at high-pressure allows employing the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres in visual optical manometry, which will be discussed in the following part.

Figure 4d illustrates the normalized emission spectra of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres at different pressures. Due to technical reasons (availability of a high-energy, focusable UV light) the sample was excited at 284 nm, which is

in the absorption range of designed compounds. When the sample is compressed to at least up to $\approx 7.35 \text{ GPa}$, the emission band of the sample shows a significant monotonous red-shift. The increased crystal-field strength and nephelauxetic effect triggered by pressure increase are primarily responsible for the observed spectral shift. As pressure increases, the interionic distances ($\text{Eu}^{2+}\text{-O}^{2-}$ and $\text{Eu}^{2+}\text{-Cl}^-$) decrease, which contributes to a stronger interaction between the Eu^{2+} ion and its ligands, promoting crystal-field splitting of the energy levels, as well as enhancing the covalent character of the bonds. Moreover, the enhancement of the nephelauxetic effect (stronger overlapping of the ligand orbitals and the ions) results in an overall contraction of the $4f$ and $5d$ electronic configurations, and a decrease in the energy separation between the ground and excited states.^[16] As the crystal-field splitting is inversely proportional to the bond length in the system ($D_q \propto 1/R^5$), the material compressed under high-pressure conditions exhibits enhanced band splitting, as the average bond lengths are shortened in the compressed system. As mentioned above, the bond length shortens with increasing pressure (see **Figure 6c,d**), which leads to the increasing D_q value and enhanced crystal-field splitting. The overall contraction of the $4f$ ground and $5d$ excited configurations, and the resulting changes in the energies of the corresponding states, may also lead to the increase of the Stokes shift with pressure, contributing to the observed red-shift of the emission band centroids. Therefore, a significant pressure-induced red-shift is observed in the emission band of the studied material. It should be noted, that the observed pressure-induced spectral shifts are fully reversible, which was confirmed by the decompression measurements (pressure release), and this behavior is crucial for the development of reliable pressure sensing materials. It is noteworthy that the bandwidth of the Eu^{2+} emission increases with increasing pressure. **Figure S4** (Supporting Information) shows the full width half maximum (FWHM) as a function of pressure. Clearly, the FWHM of the

Eu^{2+} band shows a monotonic increasing trend over the measured pressure range. The broadening of the emission band is related to several factors, such as larger crystal-field splitting of the excited states, enhanced spin–orbit coupling, and increased nephelauxetic effect and electron-phonon coupling with pressure.

Figure 4e displays the centroid of the emission band of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$ microspheres as a function of pressure, where its value is shifted from ≈ 497.3 nm ($20\,107\text{ cm}^{-1}$) to 568.8 nm ($17\,580\text{ cm}^{-1}$) when manipulating the pressure from ambient conditions up to 7.35 GPa. Note that, the relation between emission centroid and pressure is linear, which fits perfectly as: $\lambda = 9.69P + 495.06$ ($R^2 = 0.995$). Consequently, the pressure sensitivity ($d\lambda/dP$), which corresponds to the rate of spectral shift of the emission centroid with pressure, reaches up to 9.69 nm/GPa ($\approx 343.83\text{ cm}^{-1}/\text{GPa}$). Finally, pressure cycling experiments were performed to explore the pressure stability of the developed microspheres, as well as the reliability of the designed luminescent manometer and repeatability of optical pressure readouts (Figure 4f). It is shown that the emission band returns to its initial spectral position and shape after several independent compression–decompression cycles, revealing that the designed optical manometers have good stability and sensing performance. These results indicate that the pressure-induced giant spectral shift of the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres make them promising candidates for ultrasensitive, visual, and optical manometry.

2.3. Structural Stability at High-Pressure

To definitively clarify the pressure stability of the developed sensor, photoluminescence measurement as a function of pressure up to 20.67 GPa have been performed, as well as ab initio calculations to study the crystal structure, and mechanical and vibrational properties of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ under pressure. As shown in Figure 5a, it is clear that, when pressure reaches values above 8 GPa, the emission peak of $5d-4f$ transition of Eu^{2+} shows different spectroscopic behavior, compared to the constant red-shift observed for pressures below 8 GPa. When the pressure is above 9.21 GPa, the emission band starts to become asymmetric and a blue-shift is observed. As presented in Figure 5b, the emission band centroid of Eu^{2+} presents a sigmoidal dependence, i.e., the emission band centroid is ≈ 620 nm and then it starts to undergo a spectral blue-shift when the pressure reaches 12.78 GPa.

The ab initio study of tetragonal $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ was performed up to 12.30 GPa. The evolution of the unit cell parameters as a function of pressure is displayed in Figure 5c. All the cell parameters decrease with increasing pressure. The three axes show similar compressibility, indicating that the compression of the structure is isotropic under hydrostatic pressure. The pressure evolution of the unit cell volume and the polyhedral units (see Figure 5d; Figure S5a,b, Supporting Information) show a decreasing trend without volume discontinuity during compression, and is dominated by the softer polyhedral unit, in

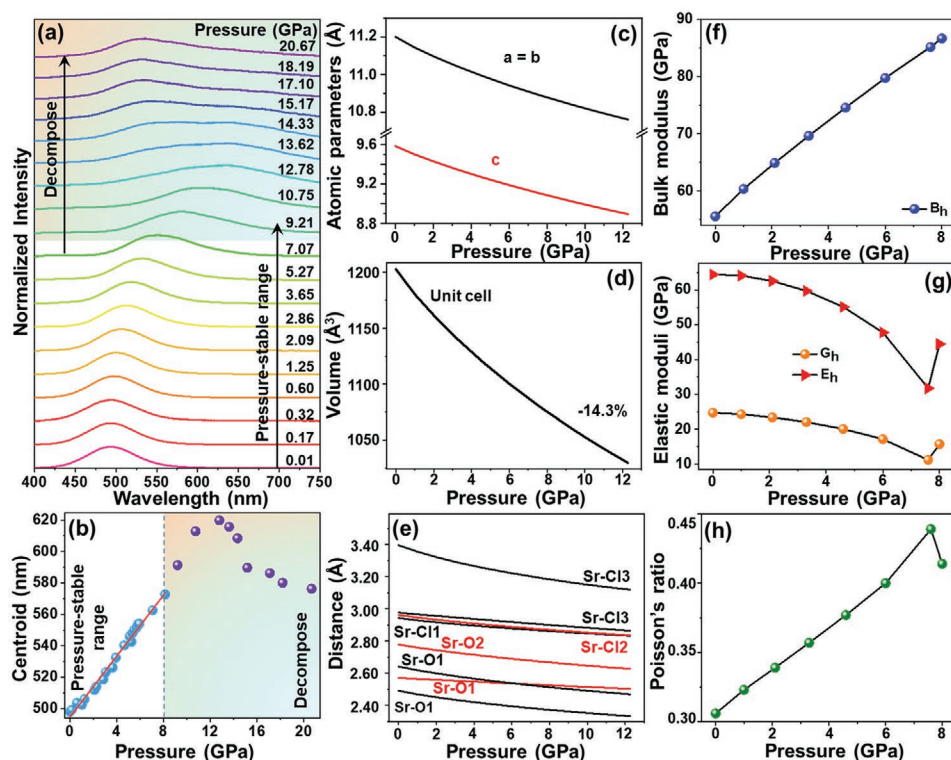


Figure 5. a) Emission band centroid of Eu^{2+} in microspheres as a function of pressure up to 20.67 GPa. b) Emission band centroid of Eu^{2+} in microspheres as a function of pressure up to 20.67 GPa. Pressure dependence of c) cell parameters and d) unit cell volume of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ based on ab initio calculations. e) Evolution of atomic bonds in the polyhedral units based on ab initio calculations. Pressure dependences of f) B , g) G , E , and h) ν parameters in $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ according to ab initio calculations (using Hill's approximation).

this case, the strontium octahedron, which decreases its volume by $\approx 14.3\%$ over the range of studied pressures. This is a consequence of the continuous decrease of the bond lengths (Si–O, Sr–O, and Sr–Cl) upon compression (see Figure 5e; Figure S6, Supporting Information).

Elastic stiffness also allows obtaining the main elastic properties of a material, which are described by the bulk modulus (B), the isotropic shear modulus (G), Young's modulus (E), the B/G relation, and Poisson's ratio^[34–36] by analytical expressions,^[37] using Hill's approximation.^[38] The pressure dependences of the bulk modulus (B), shear modulus (G), Young modulus (E), elastic moduli, and Poisson's ratio (ν) for $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ are shown in Figure 5f–h, and the pressure dependence of the B/G ratio is illustrated in Figure S7 (Supporting Information). In Figure 5f,g, the B increases monotonously and rapidly with pressure, while the G and E decrease rapidly when pressure is below 7.6 GPa and then show an abrupt increase that seems to be related to the mechanical instability above this pressure. The same inflection points at 7.6 GPa are also observed in the B/G ratios and ν . Namely, B/G ratios and ν corroborate this change in behavior. They reach their highest value (7.628 for B/G and 0.439 for ν) at 7.6 GPa, then start to decrease. It is noteworthy that when pressure is adjusted in the range of 0–8 GPa, the B/G ratios is above 1.75, which indicates that the material is ductile.^[39] The results concerning the mechanical instability at 7.6 GPa and the dynamical instability at 9.3 GPa, which agree well with the spectroscopic behavior under pressure, further confirm that $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ remains stable in the tetragonal phase up to ≈ 8 GPa. On the other hand, according to the materials Project,^[40] $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ can decompose as $\text{SrSiO}_3 + \text{SrCl}_2$. Therefore, we performed ab initio simulations of different phases of SrSiO_3 and SrCl_2 .^[41–43] Analysis of the variation of the enthalpy difference with pressure (see Figure S8, Supporting Information) shows that the material under study decomposes when pressure is higher than 9.1 GPa. Ab initio results reveal that $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ decomposes in SrSiO_3 with a cubic space

group (SG) 221 and SrCl_2 with an orthorhombic structure (SG 62) when pressure exceeds 9.1 GPa.

The ab initio calculated electronic band structures along the high symmetry directions in the first Brillouin zone (BZ) at ambient pressure is shown in Figure 6a. The band structure shows a typical semiconductor-like distribution. The valence bands are very flat with little dispersion, which agrees with a large ionic nature of the chemical bonds in $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ that leads to highly localized and polarized states in real space and, consequently, no covalent and extended bond formation within the periodic structure of the crystal. The bandgap is indirect with the minimum of the conduction band (CB) at the Γ point, while the maximum of the valence band (VB) is located at a point along the $\Gamma - X$ path that does not change with pressure. At zero pressure, the electronic band gap is 5.26 eV and increases with increasing pressure (Figure 6b). It should be noted that, in general, the observed trend with pressure of an indirect band gap is the opposite.

As for the lattice dynamic, the calculated phonon dispersion spectrum along high-symmetry directions of the BZ, at zero pressure and at 9.3 GPa, are displayed in Figure 6c,d, respectively. For low frequencies, the bands are relatively flat and no imaginary frequencies are observed. This fact reflects that the material is dynamically stable at this pressure. As shown in Figure 6c, there is a higher density of vibrational modes in the region between 0 and 250 cm^{-1} . To understand the contribution of each atomic species to the vibrational spectrum, the total (DOS) and partial (or projected onto each atom) density of states (PDOS) were calculated at zero pressure (see Figure 6e). Cl and Sr atoms contribute mainly in the region of lower frequencies up to 200 cm^{-1} , where there is a great concentration of modes, and with a minor contribution $\approx 350 \text{ cm}^{-1}$. On the other hand, Si and O atoms contribute to the entire vibrational spectrum. In particular, O atoms have an important contribution in the medium and high regions. The phonon dispersion at 9.3 GPa, displayed in Figure 6e, shows some imaginary

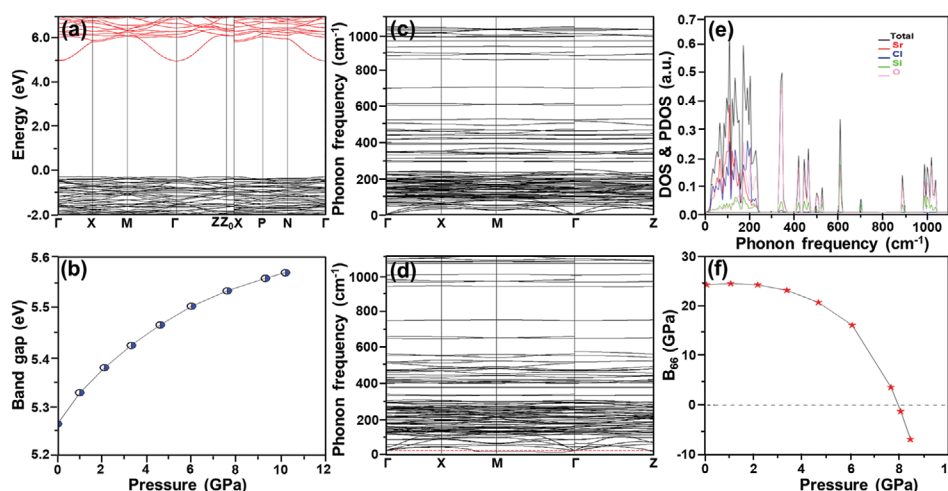


Figure 6. a) Electronic band structure of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ at ambient pressure based on ab initio calculations. b) Pressure evolutions of the electronic bandgap, according to ab initio calculations. c) Ab initio phonon dispersion at 0 GPa. d) Ab initio phonon dispersion at 9.3 GPa. e) The theoretical total one-phonon density of states in black and the partial atomic contributions of Sr (red), Cl (blue), Si (green), and O (pink) to the one-phonon density of states in $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$. f) Elastic stiffness coefficients of B_{66} in $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ as a function of pressure according to ab initio calculations.

frequencies. Therefore, the compound $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ is dynamically unstable at pressures beyond 9.3 GPa.

Regarding the elastic properties and mechanical stability of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$, this compound belongs to the tetragonal group and has seven independent elastic constants, C_{ij} . A crystal is mechanically stable at zero pressure only if the Born stability criteria are fulfilled.^[32] In the case of a tetragonal system, there are five conditions:^[33]

$$\begin{aligned} \text{(I)} & C_{11} - C_{12} > 0; \text{(II)} 2(C_{13})^2 < C_{33}(C_{11} + C_{12}); \text{(III)} C_{11} > 0; \\ \text{(IV)} & C_{44} > 0; \text{(V)} C_{66} > 0 \end{aligned} \quad (2)$$

As expected, these conditions are satisfied by the tetragonal $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ at ambient pressure. However, when a hydrostatic pressure, P , is applied to a crystal, the elastic constant, C_{ij} , must be replaced by the elastic stiffness coefficients, B_{ij} , in the above stability conditions (I to V) to obtain the “generalized Born stability criteria”. The elastic stiffness coefficients are defined as:^[34]

$$B_{ii} = C_{ii} - P, \text{ for } i = 1 \text{ to } 6 \quad (3)$$

$$B_{ij} = C_{ij} + P, \text{ for } i \neq j \text{ and } i, j = 1 \text{ to } 3 \quad (4)$$

$$B_{ij} = C_{ij}, \text{ for } i \neq j \text{ and } i, j = 4 \text{ to } 6 \quad (5)$$

with C_{ii} being the elastic constants evaluated at the current stressed state. Note that, B_{ij} reduces to C_{ij} when zero pressure is applied. In the case of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$, Figure 6f shows that the B_{66} elastic stiffness coefficient becomes negative when the pressure is above 7.9 GPa. Therefore, the “generalized stability criteria” corresponding to the (V) generalized Born stability criteria, $B_{66} > 0$, is violated. Hence, tetragonal $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ is not mechanically stable at pressure >7.9 GPa.

In order to experimentally investigate the impact of pressure on the structural properties of the investigated materials, and

correlate them with the results of theoretical calculations, we have performed XRD measurements under high-pressure conditions (up to ≈ 12 GPa). Figure 7a presents the powder XRD patterns recorded at different pressure values, clearly revealing broadening of the reflexes and their shift toward higher values of 2θ angle. Note, location of the reflexes at totally different diffraction angles 2θ , compared to the XRD data from ambient pressure (Figure 1), is simply due to the use of high-energy Mo $K\alpha$ radiation ($\lambda = 0.7107 \text{ \AA}$) for the high-pressure XRD measurements. The observed broadening effect originates mainly from the pressure-induced strains, as well as it may be related to the formation of some crystal defects in the compressed structure. Whereas, the observed shift is associated with pressure-induced contraction of the unit cell, which can be clearly seen in Figure 7b–d, showing the determined unit cell parameters ($a = b$; c) and its volume (V). Importantly, due to the high compressibility of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ host lattice, its unit cell volume decreases from ≈ 1200 to $1075 \text{ \AA}^3$, by increasing the pressure from 0.1 MPa (ambient conditions) to 12 GPa, respectively, resulting in experimentally estimated bulk modulus (B_0) at ambient condition of ≈ 74 GPa. Hence, the material compression leads to the relative decrease of the cell volume by $\approx 11\%$ around 12 GPa. Additionally, we compared and plotted in Figure 7e the evolutions of the experimentally determined and simulated V values as a function of pressure (normalized to 100% at 0.1 MPa). Note, that the performed theoretical simulations resulted in even higher compressibility for the bulk $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ material, overestimating contraction of the cell volume, especially in the high-pressure range. This effect can be associated with the observed phase transition, i.e., decomposition of the $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ crystal structure above ≈ 8 GPa, which is due to its mechanical and dynamic instabilities above ≈ 7.6 and 9.3 GPa, respectively, as was already discussed in the previous sections. The decomposition effect can be also clearly seen in the recorded XRD patterns (Figure 7a), manifested by huge

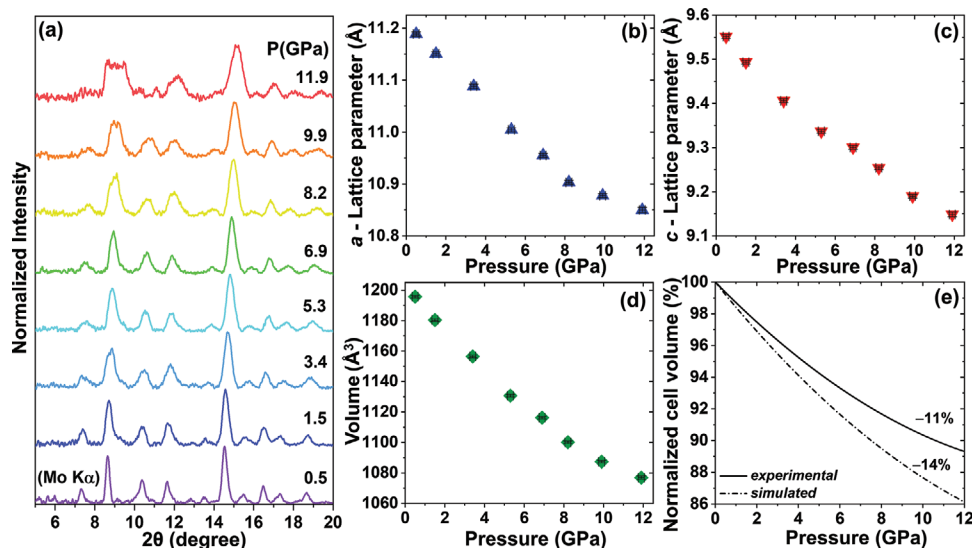


Figure 7. a) Powder XRD patterns for the investigated microspheres measured under high-pressure conditions in a DAC, using Mo $K\alpha$ radiation ($\lambda = 0.7107 \text{ \AA}$). (b–d) Experimentally determined unit cell parameters, i.e., $a = b$ (b), c (c) and V (d) as a function of pressure. e) Evolution of the relative unit cell volume as a function of pressure, based on the experimental and simulations data.

reflexes broadening, significant decrease of the signal-to-noise ratio, as well as disappearing of some low-intensity reflexes at extreme pressure values. It is worth noting that the observed structural changes are gradual, which agrees alike with the performed simulations and experimental spectroscopic data.

To confirm the decomposing process of the resultant microspheres starting at ≈ 7.6 GPa, predicted via ab initio calculations, experimental Raman spectra of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8\text{:Eu}^{2+}$ microspheres at ambient condition and high-pressure are measured, as shown in Figures S9a,b and S11a (Supporting Information). At ambient condition, the Raman peaks at 348.6, 367.5, 412.8, 439.1, 549.0, 883.9, 956.2, 1020.9, and 1071.3 cm^{-1} correspond to the Raman modes B_g , E_g , A_g , A_g , A_g , B_g , A_g , E_g , and B_g , respectively (see Figure S9b, Supporting Information). It is worth noting that the experimental Raman modes agree well with theoretical Raman spectrum based on ab initio calculations (Figure S9a, Supporting Information). The theoretical Raman-active mode frequencies for the studied material as a function of pressure are shown in Figure S10 (Supporting Information). The experimental Raman spectra as a function of pressure in compression process and decompression are displayed in Figures S11a and S12 (Supporting Information), respectively. Note that, two Raman modes of the sample, centered at ≈ 890 and 1030 cm^{-1} (marked with asterisk in Figures S11a and S12, Supporting Information), overlap with the Raman peaks from diamond anvils (from DAC), hampering their monitoring with the pressure. It is evident that all of the Raman modes undergo a shift to higher frequencies with increasing pressure from 0.18 to 7.49 GPa, so energy of the phonon modes increases, which is associated with the bonds shortening in the compressed structure. Interestingly, when pressure is above 7.49 GPa, the intensities of the Raman modes start to decrease rapidly. In particular, when pressure is over 11.24 GPa, the intensity of the most significant Raman peak initially at $\approx 960 \text{ cm}^{-1}$ undergoes a fast decay and decrease. This result agrees well with the previous prediction in the theoretical part, which confirms the good structural stability below 7.49 GPa and the decomposing

process above 11.24 GPa. The Raman shift of the most intense Raman modes of the sample, i.e., the Raman modes initially located at 370, 413, 550, and 960 cm^{-1} as a function of pressure. As shown in Figure S11b (Supporting Information), these four Raman modes undergo a linear shift to the higher frequencies, with shift rate of 4.06, 2.00, 1.10, and $4.11 \text{ cm}^{-1}/\text{GPa}$, respectively. It is worth noting that, due to the increasing strains and crystal defects in the compressed material, a general deterioration of the Raman signal with pressure is observed, which is a commonly observed effect, hampering observations of the low-intensity Raman modes.

In order to compare the pressure sensing performance of the developed Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microsphere pressure sensor with the previously reported pressure sensors, the highest achievable pressure sensitivities, i.e., $|d\lambda/dP|$, are illustrated in Figure 8. In comparison, the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ optical pressure sensor shows excellent pressure sensitivity, which are ≈ 2 times higher than the previous leaders in term of pressure sensitivity, i.e., $\text{Ca}_9\text{NaZn}(\text{PO}_4)_7\text{:Eu}^{2+}$ ($d\lambda/dP = 5.21 \text{ nm/GPa}$),^[16] and it is ≈ 6 times higher than that of recently reported visual optical pressure sensor of $\text{BaLi}_2\text{Al}_2\text{Si}_2\text{N}_6\text{:Eu}^{2+}$ ($d\lambda/dP = 1.58 \text{ nm/GPa}$).^[13] In addition, it is about ≈ 27 times higher than the commonly used optical pressure sensor ruby ($d\lambda/dP = 0.365 \text{ nm/GPa}$).^[11,12] To further present the pressure sensing capacity of the resultant compounds, we have summarized various optical pressure sensors as well as their sensitivities in Table S3 (Supporting Information). Apparently, the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres exhibit the best pressure sensing performance than any other developed optical pressure sensors, suggesting that the resultant microspheres are excellent candidates for ultrasensitive pressure sensing. The results suggest that the developed optical pressure sensor are excellent candidates for optical pressure detection with very high sensitivity for different purpose, e.g., spectroscopy under high-pressure condition, crystal structural investigations, formation of novel materials under extreme conditions, etc.

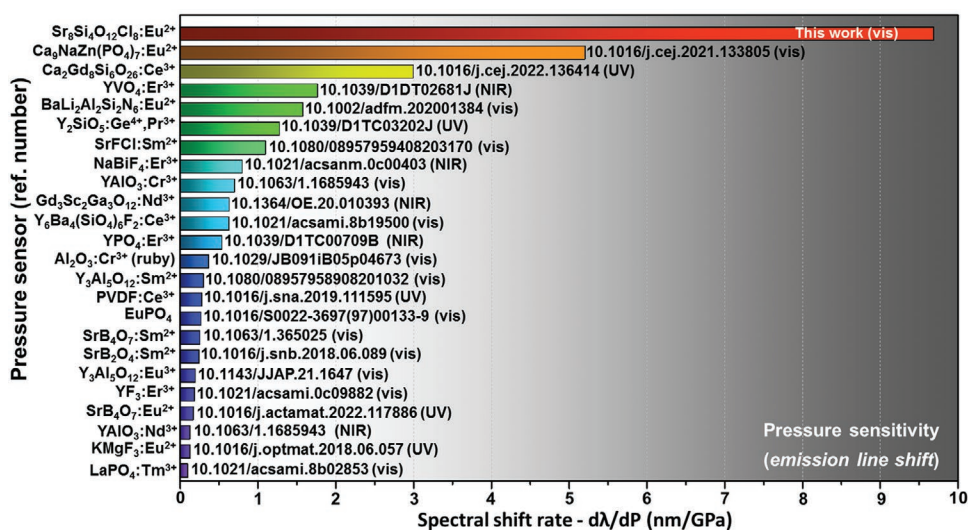


Figure 8. Pressure induced spectral shift rates ($d\lambda/dP$), i.e., pressure sensitivity, of various luminescent pressure sensors reported. The data can be found in the references (indicated by doi number).^[14,15,44–53] In all cases, the pressure sensitivity of the developed sensor in this work corresponds to the highest achievable value for all reported pressure sensor material.

3. Conclusion

In summary, a series of highly-efficient $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$ microspheres were synthesized so as to explore their feasibility for optical manometry. Excited at 350 nm, intense cyan emission originating from $4f-5d$ transition of Eu^{2+} is observed in the studied samples, and the optimal doping content is 4 mol%. With elevating the pressure up to 7.35 GPa, huge red-shift of emission band (from 497.33 to 568.82 nm) and color-tunable emissions (from cyan to green, and finally to yellow) are realized in Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres. Notably, the pressure sensitivity of the designed compound reaches up to 9.69 nm/GPa that is the highest sensitivities reported so far. Moreover, based on theoretical and experimental results, the structural stability is systematically analyzed. Considering ultra-high sensitivity and superior visual emission-color-tuning under elevated pressure, the Eu^{2+} -activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ microspheres can be applied, e.g., in accurate optical pressure sensing, detecting/prevention of the disasters of some ultra-heavy construction facilities (e.g., appearance of cracks), deep-sea exploration, as well as for the potential future high-pressure research performed under extreme conditions in planetary interiors.

4. Experimental Section

Synthesis of Eu^{2+} -Activated $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ Microspheres: By the use of high-temperature solid-state reaction method under reducing atmosphere ($\text{N}_2/\text{H}_2 = 95\%/5\%$), $\text{Sr}_{8(1-x)}\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$ (denoted as $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:8x\text{Eu}^{2+}$), $x = 0.005, 0.02, 0.04, 0.06, 0.08, 0.10$ and 0.12) microspheres, i.e., $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.04\text{Eu}^{2+}$, $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.16\text{Eu}^{2+}$, $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.32\text{Eu}^{2+}$, $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.48\text{Eu}^{2+}$, $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.64\text{Eu}^{2+}$, and $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8:0.96\text{Eu}^{2+}$, were synthesized. Herein, the raw materials of SiO_2 , SrCO_3 , NH_4Cl , and Eu_2O_3 were all bought from Aladdin Company, in which their purities were 99.99%, 99.95%, 99.5%, and 99.99%, respectively. The stoichiometric powders of SiO_2 , SrCO_3 , NH_4Cl , and Eu_2O_3 were weighted and ground for 30 min by an agate mortar. Then, they were kept in alumina crucibles and sintered at 900°C for 4 h, where the heating rate was 3°C min^{-1} . After finishing the sintering process, the final powders were collected for further characterization.

Sample Characterization: The phase compositions, surface behaviors and morphology of final products were characterized by means of an X-ray diffractometer (Bruker D8 Advance, using $\text{Cu K}\alpha_1$ radiation; $\lambda = 1.5406 \text{ \AA}$), FT-IR spectrophotometer (Bruker Tensor 27) and field-emission SEM (HITACHI SU3500) attached with an EDS analyzer, respectively. The excitation and emission spectra at ambient conditions were recorded using a fluorescence spectrometer—Edinburgh FS5. Whereas, the luminescence decay curves and quantum yield of the synthesized microspheres were measured with a fluorescence spectrometer—Edinburgh FLS1000.

To detect the emission spectra under high-pressure conditions, a small diamond anvil cell (mini-DAC), designed at the University of Paderborn (Germany) was used, equipped with ultra-low Raman fluorescence diamond anvils (IIa type) and a stainless-steel gasket of 200 μm thick. The metal gasket was pre-indented down to $\approx 70 \mu\text{m}$, and then drilled to make a center hole of $\approx 150 \mu\text{m}$ size. Four metal screws were gently tightened to precisely adjust pressure values in the DAC chamber. In order to achieve hydrostatic conditions, a pressure transmitting medium (PTM), which was composed of the mixture of methanol/ethanol/water (volume ratio of 16:3:1) was applied. The resultant microspheres, micron-sized ruby ball ($< 10 \mu\text{m}$ in diameter) and PTM were loaded into DAC chamber. Herein, a spectrometer (Andor Shamrock 500i) equipped with a silicon CCD camera (iDus) detector was applied to record the pressure-dependent emission spectra. Whereas, a focused 280 nm UV diode was used as excitation source. All luminescence spectra were corrected to

account for the apparatus response of the system. The details of the high-pressure XRD and Raman measurements are given in the SI data.

Calculation Overview: Ab initio simulations of bulk $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ were performed within the framework of DFT,^[54] as implemented in the Vienna Simulation package (VASP).^[55] The atomic species were described employing the projector-augmented wave pseudopotential (PAW).^[56] Due to the presence of oxygen atoms, to obtain accurate results a plane-wave energy cut-off of 540 eV was used. The exchange-correlation energy was described within the generalized gradient approximation (GGA) with the Armiento and Mattsson, AM05, prescription.^[57] Integrations over the Brillouin zone (BZ) were carried out with $(6 \times 6 \times 2)$ meshes of Monkhorst-Pack k-points.^[58] This procedure allows to achieve convergence in the energy better than 1 meV per formula unit. For a set of selected volumes, the cell parameters and atomic positions were fully optimized by calculating the forces on atoms and the stress tensor. In the resulting optimized configurations, the forces on atoms were less than 0.002 eV $\text{\AA}^{-1}$ and the deviation of stress tensor components from the diagonal hydrostatic form lower than 0.1 GPa. Electronic band structure calculations were carried out choosing the k-path with the Seek-path tool. The bands structure analysis was performed with the sumo package.^[59]

Lattice-dynamic calculations of the phonon modes were carried out under pressure at the zone center (Γ -point) of the BZ with the direct force-constant approach using the Phonopy package.^[60] These calculations provide the frequency of the normal modes, their symmetry, and their polarization vectors. The phonon dispersion and the projected phonon density of states (DOS) were obtained with the supercell method employing a $(2 \times 2 \times 2)$ supercell. The elastic constants were evaluated with the method implemented in the VASP code^[61] in order to study the mechanical stability and elastic properties of $\text{Sr}_8\text{Si}_4\text{O}_{12}\text{Cl}_8$ via the elastic moduli obtained from the calculated elastic stiffness constants.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by Fundamental Research Funds for the Provincial Universities of Zhejiang (SJLZ2020004), Natural Science Foundation of Ningbo (2021J063) and K. C. Wong Magna Fund in Ningbo University. T.Z. acknowledge the support from grant No. POWR.03.02.00–00-i020/17 financed by European Union through the European Social Fund under the Operational Program Knowledge Education Development. This work was supported by the Polish National Science Centre 2016/23/D/ST4/00296. This research was also supported by Ministerio de Ciencia e Innovación (MICIIN) under the National Program of Sciences and Technological Materials (PID2019-106383GB-C43/44, DOI: 10.13039/501100011033, RTI2018-101020-B-I00), by Agencia Canaria de Investigación, Innovación y Sociedad de la Información (ProID2020010067 and ProID2021010102), and by EU-FEDER funds. M.R. acknowledges support from Fondo Social Europeo and Agencia Estatal de Investigación (RYC2020-028778-I/I AEI/10.13039/501100011033).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

diamond anvil cell, lanthanides spectroscopy under pressure, optical (luminescence) manometry, rare-earth ions, remote pressure sensors

Received: December 15, 2022

Revised: February 28, 2023

Published online:

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